

FVCOM-MP: Mass Conservation Bug Diagnosis

Delaware River Estuary Test Cases

WaterPACT FVCOM-MP Development

May 2026

Delaware River Test Cases

	Case a	Case b	Case c
Role	<i>baseline (published)</i>	sensitivity b	sensitivity c
PLAST_FLOC	F	T	T
SEDIMENT_MODEL	F [†]	T	T
SED_SD50	—	0.1 mm	1.0 mm
MP water field(s)	<i>c</i>	$c_a + c_d$	
Agg. gate $d_{MP}^{\max}$	—	34 μm (excl.)	162 μm (act.)

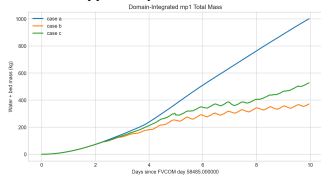
c = disaggregated conc.

c_a = aggregated, c_d = disaggregated (floc path)

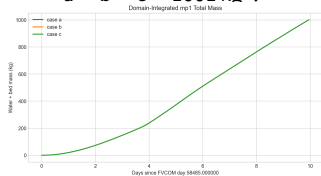
[†]sed. initialized; adv_scal/FCT bypassed • SOURCE_AG_RATE = 0 $\Rightarrow c_a \equiv 0$ • $\rho_p = 950 \text{ kg m}^{-3}$, $d_p = 100 \mu\text{m}$

Mass Conservation Progress: v001 → v006 → v008 → v009

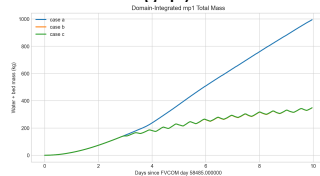
v001 — original (full sediment active)



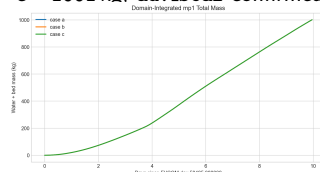
v008 — Fix B+; sed adv bypassed
 $a = b = c = 1001 \text{ kg}$ ✓



v006 — sed adv/FCT bypassed; NaN isolated (650 kg gap)



v009 — Fix B+ + adv_scal active (no backward/FCT)
 $a = b = c = 1001 \text{ kg}$; adv_scal confirmed safe ✓



Bug 1: adv_scal_backward/fct_sed corrupt shared PFPXB/PFPYB arrays

→ disable backward scheme; adv_scal is safe

Bug 2: rho_water = NaN → eflx = NaN → neg-clamp destroys mass

→ isnan sentinel on eflx/nflx

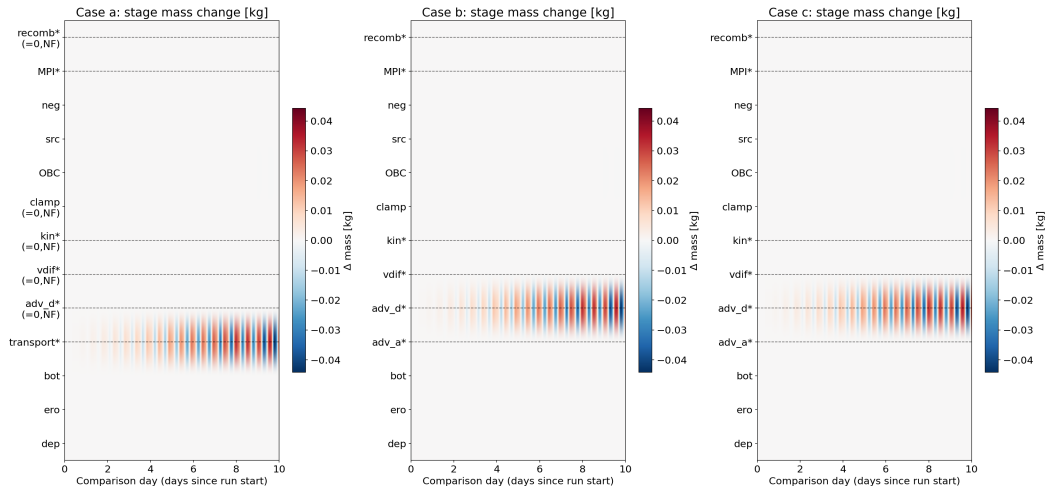
In-Run Mass Checkpoint Framework

CP	Stage	ΔM expected
0	start of timestep	<i>reference</i>
1–3	dep / ero / bed update	$\Delta(\text{water} + \text{bed}) = 0$
3p5	adv(conc_a) only	0* (conc_a = 0 now)
4	adv(conc_d)	OBC tidal exchange
5	vdif	0*
6	kinetics	0* (conc_a = 0 now)
7	upper clamp	0
8	OBC	0 (absorbed in adv)
9	PTsource (river)	river import
10	neg-clamp	0 (loss only)
11	MPI halo sync	0*
12	end of timestep	verify $\approx$ CP0

* dashed row in heatmap — must be exactly zero

Stage-Delta Heatmap: Fix B+ Active, All Cases, 10-day Run

Stage mass-change heatmap (diverging: red = gain, blue = loss)
Dashed lines = stages that must be exactly conservative



Grand-Total Mass Timeseries: CP0 vs. CP12 per Step

